

Language Instantiation and Cross-Language Equivalence in Dimensional Human Field Theory (DHFT)

DHFT Technical Note

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Abstract

Language instantiation in Dimensional Human Field Theory (DHFT) establishes the conditions under which the system may be expressed across multiple languages while preserving structural invariance.

Cross-language equivalence is enforced such that all language instantiations remain structurally identical representations of the same system.

Language instantiation governs representation across languages and does not define system structure, admissibility, or system use conditions.

I. Field Context

Dimensional Human Field Theory (DHFT) is a constraint-based system defined over invariant variables:

- Load (L)
- Capacity (C)
- Boundary (B)

These variables and their structural relations are defined in canonical DHFT sources and are not modified across language instantiations.

II. Position in System Architecture

Language instantiation applies only after admissible entry into DHFT.

Dependency order:

- Structural Stability Science (SSS) — field conditions
- Structural Identification Method (SIM) — derivation conditions
- DHFT — system definition
- Admissibility Conditions — validity constraints
- Entry Protocol — access conditions
- Boundary of Use — application constraints
- Interface Protocol — output constraints
- Language Instantiation — cross-language representation constraints

Language instantiation does not define, override, or modify upstream conditions. It enforces structural equivalence across language representations.

III. Cross-Language Equivalence Constraint

All language instantiations of DHFT must preserve structural equivalence.

Structural equivalence requires:

- invariant variable identity (L , C , B)
- invariant variable definitions
- invariant structural relations
- invariant admissibility conditions

Variable notation (L , C , B) remains invariant across all language instantiations.

Structural equivalence across languages does not imply semantic equivalence outside the system.

Language variation does not introduce or remove system meaning.

IV. Translation Constraint

Translation does not modify system structure.

Language instantiation:

- does not introduce new variables
- does not alter variable definitions
- does not alter structural relations

Any translation that modifies structure constitutes non-admissible representation.

Partial or fragmentary instantiations do not constitute admissible DHFT representations.

Conditions governing language instantiation are defined in *Language Instantiation and Cross-Language Equivalence in DHFT*.

V. Non-Equivalence Constraint

No term in a target language is treated as equivalent unless it preserves:

- structural definition
- system function
- relational position within the system

Apparent semantic similarity does not constitute structural equivalence.

VI. Drift Condition

The following indicate non-admissible language instantiation:

- substitution of variables with domain-specific terminology
- introduction of interpretive, psychological, or narrative constructs
- alteration of variable definitions
- alteration of structural relations

Any such condition constitutes structural drift.

Bidirectional translation must preserve structural equivalence. Violation constitutes non-admissible instantiation.

VII. Canonical Equivalence Statement

All language instantiations of DHFT are structurally equivalent.

No language version holds interpretive priority.

All admissible instantiations represent the same system.

VIII. Interpretation Boundary

Meaning is the exclusive domain of the human agent.

DHFT does not generate, encode, or transmit meaning.

Attribution of meaning occurs outside the system and does not constitute system structure.

IX. Enforcement Statement

All admissible language instantiations:

- preserve structural invariance
- maintain reduction to (L) , (C) , and (B)
- remain consistent with canonical definitions

Instantiations that violate these conditions:

- are non-admissible
- do not constitute admissible DHFT representation

IX. Canonical Language Instantiation

Canonical DHFT may exist in multiple language instantiations.

Each instantiation preserves invariant variables, structural relations, and admissibility conditions.

Canonical language instantiations are structurally equivalent and non-hierarchical.

No language version holds interpretive or structural priority.

All designated canonical instantiations constitute DHFT.

X. System Reference

The system is specified in:

Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System

DOI: 10.5281/zenodo.19075948

All valid system descriptions reduce to Load (*L*), Capacity (*C*), and Boundary (*B*).

No additional variables are introduced at the minimal layer.

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This work defines original structural models and associated terminology within Dimensional Human Field Theory (DHFT).

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